Sotiriou, Spontaneous scalarization in generalized scalar-tensor theory, Phys. Rev. **D99**, 124022 (2019), arXiv:1904.06365 [gr-qc].

[43] E. S. Demirboğa, Y. E. Şahin, and F. M. Ramazanoğlu, Subtleties in constraining gravity theories with mass-radius data, Phys. Rev. D **108**, 024028 (2023), arXiv:2303.01910 [gr-qc].

[44] B. D. Metzger, L. Hui, and M. Cantiello, Fragmentation in Gravitationally Unstable Collapsar Disks and Subsolar Neutron Star Mergers, Astrophys. J. Lett. **971**, L34 (2024), arXiv:2407.07955 [astro-ph.HE].

[45] B. C. Stephens, W. E. East, and F. Pretorius, Eccentric Black Hole-Neutron Star Mergers, Astrophys. J. Lett. **737**, L5 (2011), arXiv:1105.3175 [astro-ph.HE].

[46] W. E. East, F. Pretorius, and B. C. Stephens, Eccentric black hole-neutron star mergers: effects of black hole spin and equation of state, Phys. Rev. D **85**, 124009 (2012), arXiv:1111.3055 [astro-ph.HE].

[47] J. a. V. M. Muniz and R. F. P. Mendes, On the phase transition mechanism of spontaneous scalarization (2025), arXiv:2503.11385 [gr-qc].

End Matter

Computing the equilibrium NS solutions: A static, spherically-symmetric metric can be put in the form

$$g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = -e^{\nu(r)}\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1-2\mu(r)/r} + r^2\mathrm{d}\Omega^2, \quad \text{(E1)}$$

NS structures in this case can be obtained by solving the modified Tolman–Oppenheimer–Volkoff (TOV) equations. Details can be found in Tuna *et al.* [29].

One needs to know the nuclear matter equation of state (EOS) to solve the TOV equations. A simple choice is a polytrope that relates the pressure of nuclear matter to the *rest mass density* of baryons $\tilde{\rho}_\mathrm{r}$ (in the Jordan frame, i.e., in the frame of the metric $\tilde{g}_{\mu\nu}$)

$$\tilde{\rho} = \tilde{\rho}_\mathrm{r} + \frac{C}{\Gamma - 1}\tilde{\rho}_\mathrm{r}^\Gamma, \qquad \tilde{p} = C\tilde{\rho}_\mathrm{r}^\Gamma, \quad \text{(E2a)}$$

where C and Γ are constants. Note that the two equations are related by the first law of thermodynamics. This relationship is sometimes expressed in terms of the number density of baryons $\tilde{n}$, which is linearly related to the baryon rest mass density as $\tilde{\rho}_\mathrm{r} = m_\mathrm{b}\tilde{n}$. In this study, we employ the slightly more complicated *piecewise polytropic* EOS introduced in Read *et al.* [30].

Once the TOV equations are solved, the total energy of the spacetime, the Arnowitt-Deser-Misner (ADM) mass M_ADM, is given by

$$M_\mathrm{ADM} = \mu(r \to \infty). \quad \text{(E3)}$$

Baryon mass, which is the integral of $\tilde{\rho}_\mathrm{r}$ over the proper volume of the star ($r < R$) in the Jordan frame is

$$M_\mathrm{b} = m_\mathrm{b}N_\mathrm{b} = \int_0^R 4\pi r^2 \tilde{\rho}_\mathrm{r} A^3 \left(1 - \frac{2\mu}{r}\right)^{-1/2} \mathrm{d}r. \quad \text{(E4)}$$

This is the total energy we would have if we separated all the N_b baryons in the star into distant places in a thought experiment.

We repeat the above procedure many times to obtain Fig. 3. In a similar vein, we plot the radius and central density of the stars (in the Jordan frame) in Fig. E1. The differences $\Delta\tilde{R}$ and $\Delta\tilde{\rho}_\mathrm{c}$ can be quite prominent. This was an important factor in constraining scalarization via mass-radius observations of NSs in Tuna *et al.* [29].

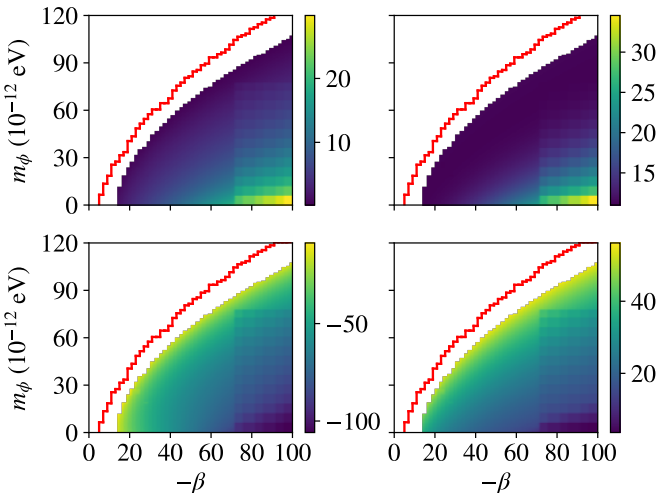

FIG. E1. Analog of Fig. 3 for different stellar parameters. Upper row: $\Delta\tilde{R}$, the radius difference between the two stars which have the maximum ADM mass difference for the same baryon mass (left) and the stellar radius at the scalarization branch-off point $\tilde{R}$ (right) in km. Lower row: Similar to upper row, but for the central energy density difference $\Delta\tilde{\rho}_\mathrm{c}$ (left) and $\tilde{\rho}_\mathrm{c}$ at the scalarization branch-off point (right) in $10^{16}\,\mathrm{kg/m^3}$.

Our relaxation scheme sometimes fails to converge to a solution [29], and we repeat the procedure with slightly different initial guesses. There might still be failures after many trials, which can be easily filled with interpolation on the (β, m_ϕ) plane. We obtained all the data points eventually, hence did not need interpolation, but this is likely not an optimal use of computational resources. We provide an earlier, incomplete version of Fig. 3, Fig. E2, as an example of the possible need for interpolation.

Basic aspects of first-order phase transitions: Here we summarize how the Landau ansatz of Eq. (3) explains first-order scalarization as seen in Fig. 4. To start with, for $\chi > 1/4$, the only equilibrium point and the global minimum of Eq. (3) is at $Q = 0$. The only configuration for a given M_b is stable and unscalarized, and we are in the region to the left of the line A in Fig. 2.

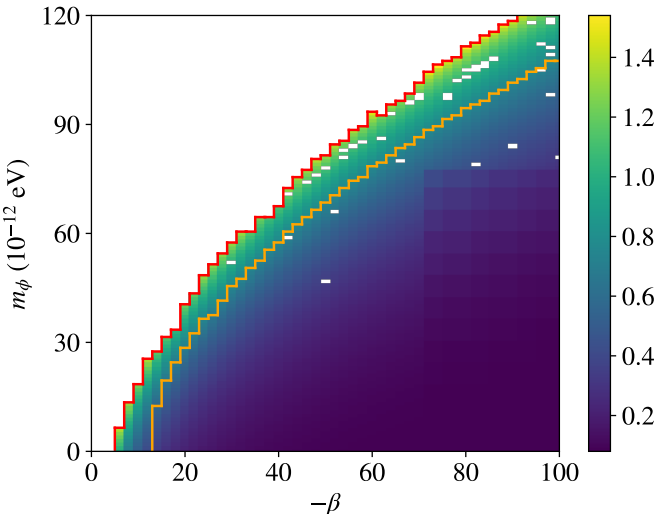

FIG. E2. An earlier version of the right subplot of Fig. 3, where we also give the ADM mass everywhere. Our code failed on the scattered white dots at this stage, which could be filled in by interpolation, but ultimately this was not needed since we had the resources to obtain the solutions eventually. The orange line separates first- and second-order scalarization, the former being characterized by $\Delta M > 10^{-5} M_\odot$.

As M_b increases, χ drops below 1/4 and local maxima ($-$) and outer local minima ($+$) appear at

$$Q_\pm^2 = [-b/(2c)] \left(1 \pm \sqrt{1 - 4\, ac/b^2}\,\right). \tag{E5}$$

However, the *global* minimum is still located at $Q = 0$ if $\chi > 3/16$. This is the case where the unscalarized ($Q = 0$) solution is the globally stable one, and the stable scalarized solutions are metastable. In Fig. 2, this corresponds to the region between vertical lines A and B. The local maxima are also equilibrium solutions, they are the unstable scalarized configurations (dotted line in Fig. 2) which have the highest total energy, i.e., the lowest binding energy among the equilibria.

As M_b increases further, we arrive at $\chi = 3/16$, where the energies at $Q = 0$ and the outer minima become equal. This corresponds to line B in Fig. 2, and it is the actual point of the first-order phase transition. If χ becomes less than 3/16, i.e., to the right of line B, the scalarized solution at Q_+ becomes the globally stable one, and the unscalarized stars become metastable. Note that, at the point of transition, the global minimum jumps *discontinuously* from $Q = 0$ to one of $Q = \pm Q_+$, which is a hallmark of first-order phase transitions.

Further increasing M_b brings us to $a = 0$ or $\chi = 0$, for which the maxima at $Q = \pm Q_-$ join the local minimum at $Q = 0$. This corresponds to the line C in Fig. 2. For negative values of a, hence of χ, we have a local maximum at $Q = 0$, and minima at $Q = \pm Q_+$. This corresponds to having one unstable GR solution and one stable scalarized solution, which is the region to the right of line C.

Extracting the Landau ansatz parameters: For fixed (β, m_ϕ) and EOS, let us expand each parameter in the ansatz (3) as a power series around the critical baryon mass (at the branch-off point), that is,

$$a(M_\mathrm{b}) = \quad a_1(M_\mathrm{b} - M_\mathrm{crit}) + \dots \tag{E6}$$

$$b(M_\mathrm{b}) = b_0 + b_1(M_\mathrm{b} - M_\mathrm{crit}) + \dots \tag{E7}$$

where we used the fact that $a(M_\mathrm{crit}) = 0$. Consider the scalarized equilibrium solutions of first-order scalarization ($b_0 < 0$) that are close to the branch-off point, i.e. those stars with $M_\mathrm{b} \approx M_\mathrm{crit}$. These are unstable solutions which correspond to the local maxima in Eq. (E5), and their scalar charge $Q = Q_-$ is given by

$$Q^2 = -\frac{a_1}{b_0}(M_\mathrm{crit} - M_\mathrm{b}) + \mathcal{O}\left[(M_\mathrm{b} - M_\mathrm{crit})^2\right]. \tag{E8}$$

Hence, the corresponding ADM mass is

$$M_\mathrm{ADM} - M_0 = -\frac{1}{2}\frac{a_1^2}{b_0}(M_\mathrm{b} - M_\mathrm{crit})^2 + \dots. \tag{E9}$$

Combining the last two equations in the leading order, we finally obtain

$$2(M_0 - M_\mathrm{ADM}) = b_0\, Q^4 + \dots, \tag{E10}$$

which holds well for stars with baryon masses close to M_crit. In summary, we calculate the ADM mass and the scalar charge of a star near the critical baryon mass, as well as M_0, which is the ADM mass of an unscalarized ($Q = 0$) star with the same baryon mass. If we repeat this for different baryon masses, the $(M_0 - M_\mathrm{ADM})$ versus Q^4 curve for small enough Q is simply a line with slope $b_0/2 = b(M_\mathrm{crit})/2$.

Eq. (E10) holds for second order scalarization as well, hence we can perform the same fitting procedure as we change the theory parameters (β, m_ϕ) without worrying about the order of the phase transition. Then, we can obtain b_0 as a function of (β, m_ϕ) (and the EOS), whose sign determines the order of scalarization we have on different parts of the STT parameter space. We reported these findings in the main text, see Fig. 5.

The fitting procedure above requires a numerical setup with particularly high precision. M_ADM and M_0 are very close to each other near the critical baryon mass. Hence, when we take their difference, we suffer from the phenomenon of *loss of precision* (also known as *catastrophic cancellation*): The distinct numerical errors in M_ADM and M_0 do not cancel each other, and the relative error grows tremendously due to $|M_\mathrm{ADM} - M_0|$ being exceedingly small compared to either of the masses. We decreased the finite differencing step sizes and the tolerances of the Newton iteration in our relaxation scheme more than an order of magnitude compared to our standard computations, so that the fit is not overwhelmed by the error. See Fig. E3.

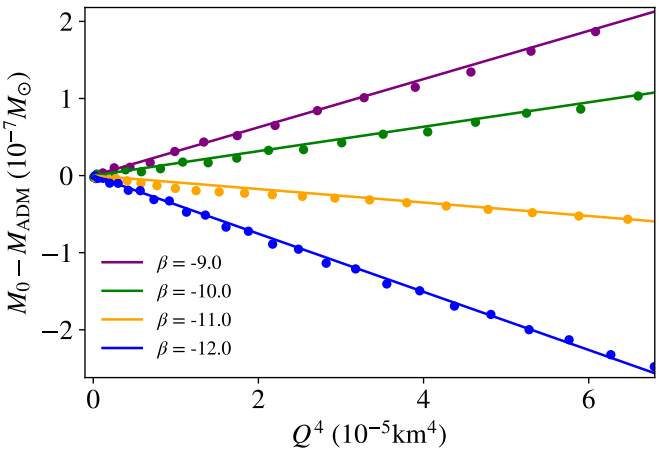

FIG. E3. Sample linear fits that yield b_0 via Eq. (E10) for various values of β ($m_\phi = 0$). The transition from second-order scalarization to first-order occurs around $\beta = -10.6$, where the slope of the fitting line, hence b_0, vanishes. See Fig. 5 for the complete curve $b_0(\beta)$. The data points have small deviations from the fit lines due to numerical errors, and we exclude the points closest to the origin from the fit due to high relative errors. There are further data points outside the frame.